Roll No.: ....

END SEMESTER EXAMINATION DECEMBER 2025

Name of the Program: B.Tech (Biotech) Semester: III

Name of the Course: Introduction to AI Course Code: TCS-542

Time: 3 Hours **Maximum Marks: 100**

Note:

i. **All questions are compulsory.**
ii. Answer any two sub questions among a, b & c in each main questions
iii. Total marks in each main question are twenty.
iv. Each questions carries 20 marks

Q1	(20Marks)	
(a)	What is an *Intelligent System*? Explain its core concepts along with examples of intelligent systems.	CO1
(b)	Discuss the Applications of AI. For education domain, describe a specific AI-based application and how it improves performance over human methods.	CO2
(c)	Design an *intelligent agent* model for an online food delivery application. Explain its anatomy, structure, and types of agents used.	CO1
Q2		
(a)	Explain the concept of a *Problem Solving Agent*. Describe the key steps in formulating a problem in AI.	CO2
(b)	Analyze the performance trade-offs between *A** and *Greedy Best-First Search* in finding optimal solutions.	CO3
(c)	(i) Differentiate between *uninformed* and *informed search methods* with suitable examples. (ii) Evaluate the role of *heuristic functions* in informed search algorithms. How do they affect efficiency and accuracy?	CO3
Q3		
(a)	(a) Differentiate between monotonic and non-monotonic logic in Artificial Intelligence. Explain why non-monotonic reasoning is required in real-world AI systems. (b) A jar contains 5 red, 7 blue, and 6 green marbles. One marble is taken out at random. Find the probability that it is: (i) Blue or green (ii) Not red (iii) Neither blue nor green	CO4 CO4 CO4
(b)	Consider the following propositions: • P: "The robot has sensors." • Q: "The robot can perceive the environment." • R: "The robot can make intelligent decisions." Formulate logical expressions for the following statements and determine their logical	

	equivalence: i) If the robot has sensors and can perceive the environment, then it can make intelligent. decisions. ii) If the robot cannot perceive the environment, then it cannot make intelligent decisions. iii) If the robot can make intelligent decisions, then it must have sensors and be able to perceive the environment.	
(c)	Compare *Semantic Networks*, *Minsky Frames*, and *Case Grammar Theory* in representing knowledge.	
Q4		
(a)	Represent the following English statements using Predicate Logic: 1. Every bird can fly. 2. Some students are not attentive. 3. All cats are mammals. 4. There exists a city where everyone loves music. 5. If an animal is a dog, then it is loyal.	CO5 CO5 CO5
(b)	Explain the basic structure and working of PROLOG as a programming language for Artificial Intelligence. Discuss its key symbols with suitable examples to illustrate how PROLOG performs logical inference.	
(c)	Why inference reasoning is needed in Artificial Intelligence. Differentiate between forward reasoning and backward reasoning with suitable examples.	
Q5		
(a)	Design a simple *Rule-Based Expert System* for diagnosing car engine problems.	CO6
(b)	Explain Minsky's Frame Theory as a method of knowledge representation in Artificial Intelligence. Using a suitable example of a "Hospital Visit Scene," illustrate how frames are used to represent real-world situations.	CO6
(c)	Critically analyze the significance of Domain Exploration in the development of Expert Systems. Discuss how an improperly defined domain can affect system accuracy and knowledge representation.	CO6